DATA CENTER @ UNIVERSITY OF MISSOURI – KANSAS CITY

CS5525 – Cloud Computing

FEBRUARY 27, 2028

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PART 1: Pre-visit Research

1. What is a data center and its primary purpose?

A data center is a centralized facility that houses computing infrastructure including servers, storage systems, and networking equipment. Its primary purpose is to store, process, manage, and distribute digital data and applications reliably and securely. Think of it as a specialized warehouse for computing resources that supports everything from university systems to cloud services.

2. Key components you'd expect to see:

- **Servers**: Physical machines running applications and storing data
- **Storage systems**: Arrays of hard drives and SSDs for data storage
- **Networking equipment**: Switches, routers, and firewalls for connectivity
- **Power infrastructure**: UPS systems, generators, and power distribution units
- **Cooling systems**: HVAC units, chillers, and air circulation systems
- **Monitoring systems**: Environmental and performance monitoring tools

3. Why cooling is critical:

Computing equipment generates significant heat during operation. Without proper cooling:

- Servers can overheat and fail, causing downtime
- Performance degrades as components throttle to prevent damage
- Hardware lifespan decreases significantly
- Energy efficiency drops

Data centers typically maintain temperatures between 64-80°F (18-27°C) with proper humidity control to ensure optimal performance and equipment longevity.

4. Security measures:

- **Physical security**: Biometric scanners, key card access, security guards
- **Perimeter protection**: Fencing, cameras, motion detectors
- **Environmental monitoring**: Fire suppression systems, leak detection
- **Access controls**: Multi-factor authentication, visitor escorts
- **Network security**: Firewalls, intrusion detection systems
- **Redundancy**: Multiple power sources, backup systems

PART 2 On Site Observation Documentation

1. Sketch Server Rack

```
|    [ VISUAL COMPONENT SKETCH ]    | [ COMPONENT DESCRIPTION & NOTES ]
===================================================================================
| ////////////////////////////////// | <-- [OVERHEAD CABLE TRAY SYSTEM] (Yellow, Top of Rack)
|    \ \ \  | | | |  / / /          |   * Bundles of Yellow Fiber cables & Blue Copper [1, 2]
===================================================================================
| [================================] | <-- [Fiber Patch Panel (Enclosure)]
| [      (Yellow Caution Label)     ] |   * Yellow fiber trunks entering from overhead [1].
| [0000 0000 0000 0000 0000 0000] | <-- [Fiber Distribution Panel] (Dense Fiber LC/SC connectors)
| [--------------------------------] |
| [0000 0000 0000 0000 0000 0000] | <-- [Primary Network Switch (1U)]
| [ [=] [=] [=] [=]  <- (Fiber SFP) ] |   * Heavily populated with Yellow Fiber cables [1].
| [--------------------------------] |
| [::::::::::::::::::::::::::::::::] | <-- [Secondary Network Switch (1U)]
| [ [=] [=] [=] [=]  <- (Blue Copper) ] |   * Connected via Blue Ethernet patch cables [1].
| [--------------------------------] |
| [________________________________] | <-- [Blanking Panel (1U)]
| [________________________________] |   * Used for airflow management [3].
| [--------------------------------] |
| [ / \ / \ / \ / \ / \ / \ / \ / \ ] | <-- [Compute Server 1: Dell EMC (2U)]
| [ \ / \ / \ / \ / \ / \ / \ / \ / ] |   * Signature Black Hexagonal Bezel [3].
| [--------------------------------] |
| [ / \ / \ / \ / \ / \ / \ / \ / \ ] | <-- [Compute Server 2: Dell EMC (2U)]
| [ \ / \ / \ / \ / \ / \ / \ / \ / ] |   * Identical unit for redundancy/clustering [3].
| [--------------------------------] |
| [ / \ / \ / \ / \ / \ / \ / \ / \ ] | <-- [Compute Server 3: Dell EMC (2U)]
| [ \ / \ / \ / \ / \ / \ / \ / \ / ] |   * Identical unit [3].
| [--------------------------------] |
| [_______[ PDU ]_________________] | <-- [Power Distribution Unit (PDU)] (PDU Labeled Panel) [3]
| [--------------------------------] |
| [________________________________] | <-- [KVM Console / Slide-out Shelf (1U)]
| [---(Pull-out Drawer Handle)----] |   * For local crash-cart access [3, 4].
| [--------------------------------] |
| [ [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] ] | <-- [Storage Array / Drive Shelf]
| [ [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] ] |   * Multi-disk chassis (e.g., Promise Storage) [3].
| [--------------------------------] |
| [ [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] ] | <-- [Additional Storage Array]
| [ [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] ] |
| [--------------------------------] |
| [________________________________] | <-- [Blanking Panel (Large)]
| [________________________________] |
| [________________________________] |
| [________________________________] |
| [________________________________] |
| [--------------------------------] |
| [ (  APC  ) [====] [====] (  )  ] | <-- [APC UPS / Backup Battery Unit]
| [================================] |   * Heavy unit installed at the base [1].
===================================================================================
|         [ FLOOR / PLENUM ]        |
```

2. HW:

a) Three specific types of hardware seen in the sources:

1. Servers:
    - Dell EMC servers mounted inside the server cabinets [1, 2]
    - Exposed Dell server chassis showing internal components resting on a yellow lift cart [3]
2. Uninterruptible Power Systems (UPS):
    - Emerson Liebert Npower unit [4]
    - Vertiv Liebert EXM unit [5]
3. Storage Arrays:
    - Rack-mounted, multi-drive storage bays with visible "PROMISE" branding installed alongside the servers in the racks [2]

```
+-----------------------+
|   Server Cabinet      |
|   [1] Dell EMC Server |
|   [2] PROMISE Storage |
+-----------------------+

+-----------------------+
|   Server Cabinet      |
|   [1] Dell EMC Server |
|   [2] PROMISE Storage |
+-----------------------+

+-----------------------+
|   Exposed Dell Server |
|   Chassis [3]         |
+-----------------------+

+-----------------------+
|   Emerson Liebert     |
|   Npower UPS [4]      |
+-----------------------+

+-----------------------+
|   Vertiv Liebert      |
|   EXM UPS [5]         |
+-----------------------+
```

b) Primary function of each piece of hardware:

1. Servers:
    - Primary high-performance computers that process user requests, host applications, and manage data computations for a network.
2. Uninterruptible Power Systems (UPS):
    - Backup power systems that provide continuous, temporary battery power to the server racks in the event of a facility power outage, allowing systems to remain online or shut down safely without data loss.
3. Storage Arrays:
    - Dedicated, high-capacity storage enclosures that hold and manage massive amounts of data, providing centralized storage that the servers rely on to read and write information.

3. SW

a) Why Operating Systems are vital to server function:

Operating systems serve as the critical interface between server hardware and applications. They manage essential functions including:

- **Resource allocation**: CPU, memory, and storage management
- **Process scheduling**: Running multiple applications simultaneously
- **Hardware abstraction**: Allowing software to interact with different hardware components
- **Security**: User authentication and access controls
- **Network communication**: Managing data transfer and connectivity

Without an OS, servers would be unable to execute programs or manage resources effectively.

b) Two common server operating systems:

Linux (e.g., Ubuntu Server, Red Hat Enterprise Linux)

- Used for: Web servers, database hosting, containerization, cloud computing
- Benefits: Open-source, highly customizable, excellent security, cost-effective
- Common in: Academic environments, web hosting, development servers

Windows Server

- Used for: Enterprise applications, Active Directory services, Microsoft-based environments
- Benefits: Integration with Microsoft ecosystem, user-friendly management tools
- Common in: Corporate environments, .NET applications, file and print services

c) & d) Essential data center management software:

Infrastructure Monitoring Software (e.g., Nagios, Zabbix, PRTG)

- Monitors server health, network performance, and resource utilization
- Provides real-time alerts for hardware failures or performance issues
- Tracks temperature, power consumption, and system availability
- Essential for preventing downtime and maintaining optimal performance

e) How data center software differs from personal computer software:

Data Center Software:

- **Scale**: Manages hundreds/thousands of servers simultaneously

- **Reliability**: Designed for 24/7 operation with minimal downtime
- **Automation**: Heavily automated for efficiency and consistency
- **Multi-tenancy**: Serves multiple users/organizations concurrently
- **Redundancy**: Built-in failover and backup capabilities

Personal Computer Software:

- **Single-user focus**: Designed for individual use
- **Interactive**: Emphasizes user interface and experience
- **Flexibility**: Allows frequent updates and customization
- **Resource usage**: Optimized for limited hardware resources

4. Cooling Systems Types Observed:

Old Data Center: Room-Level Cooling

- Uses traditional air conditioning units to cool the entire room
- Less efficient approach as it cools all spaces uniformly
- Higher energy consumption due to cooling unused areas

New Data Center: Cabinet-Level Cooling

- Targets cooling directly at equipment cabinets/racks
- More precise and energy-efficient approach
- Reduces wasted cooling on non-equipment areas

Cold Air Distribution Methods:

Traditional Room Cooling:

- Cold air is blown into the general room space
- Air circulates throughout the entire data center floor
- Less controlled airflow patterns

Modern Cabinet Cooling:

- **Hot Aisle/Cold Aisle Configuration**: Server racks are arranged so cold air intakes face one direction (cold aisle) while hot exhausts face the opposite direction (hot aisle)
- **Targeted airflow**: Cold air is directed specifically into server intake vents at the front of racks
- **Containment systems**: Some setups use physical barriers to separate hot and cold air streams, preventing mixing
- **Under-floor or overhead distribution**: Cold air may be delivered through raised floors or overhead ducts directly to equipment

Benefits of the Modern Approach:

- **Energy efficiency**: Up to 30-40% reduction in cooling costs
- **Better temperature control**: More consistent temperatures at equipment level
- **Reduced hot spots**: Prevents equipment overheating
- **Scalability**: Easier to add cooling capacity as equipment grows

This evolution reflects the industry trend toward more sustainable and cost-effective data center operations.

5. Power Infrastructure

Key power infrastructure and Uninterruptible Power Systems (UPS) in the facility:

1. Standalone UPS Units:
    - Emerson Liebert Npower (80kVA) [1]
    - Vertiv Liebert EXM [2]
2. Rack-Mounted UPS:
    - APC backup power supply [3]
3. Power Distribution Panels:
    - GE Spectra Series Power Panelboard (277/480V) [4]

Importance of backup power systems:

- Prevent data loss and corruption
- Ensure continuous uptime
- Protect server hardware from power surges, spikes, or drops
- Provide vital time for safe shutdown or generator startup in a full facility outage

Part 3: Post-Visit Reflection

Reflecting on my recent visit to a state-of-the-art data center, I am struck by the sheer complexity and scale of the operation. The experience has left me with a newfound appreciation for the intricate systems that power our digital world.

1. Summary: The key takeaway from my visit is the critical importance of backup power systems and cooling infrastructure in maintaining continuous uptime. I was surprised by the vast array of Uninterruptible Power Systems (UPS) and the meticulous attention to detail in power distribution. The facility's design ensured that every component, from servers to power panels, had multiple redundancies in place to prevent any single point of failure.
2. Sustainability: Data centers are notorious for their energy consumption and environmental impact. However, this facility demonstrated several measures to improve energy efficiency. For instance, the use of hot-aisle/cold-aisle containment optimized cooling, while the deployment of energy-efficient servers and power distribution units reduced overall energy consumption. Additionally, the data center's commitment to using

renewable energy sources and implementing waste heat recovery systems highlighted its dedication to sustainability.

3. Career Connections: The operation of a data center requires a diverse range of skilled professionals. From IT technicians and network engineers responsible for maintaining servers and ensuring seamless connectivity, to electrical engineers and HVAC specialists tasked with managing power and cooling systems, the facility offered a glimpse into various career paths. Furthermore, the growing emphasis on data security and sustainability presents opportunities for cybersecurity experts and environmental consultants in the industry.
4. Future of Data: As the demand for data storage continues to grow, the future of data centers lies in innovation and scalability. I envision a future where data centers become more modular, allowing for rapid expansion and adaptation to new technologies. Moreover, the increasing focus on sustainability will drive the development of energy-efficient cooling techniques, renewable energy integration, and waste heat utilization.

In conclusion, my visit to the data center has been an eye-opening experience, revealing the intricate systems that underpin our digital lives. The facility's commitment to sustainability and efficiency, coupled with the diverse range of career opportunities, has sparked my curiosity about the future of data storage and the role data centers will play in shaping our digital landscape.